(Please write your Enrollment Number) Enrollment No._____

MID-TERM EXAMINATION
(Course Name: B.Tech ECE-AI) (Semester: 2nd Semester)
(May, 2023)OFF LINE mode

Subject Code: BEC 106	**Subject: Signals and Systems**
Time : 1 ½Hours	**Maximum Marks : 30**

Note: Q. 1 is compulsory.

Q 1	(Attempt all the parts)	**(2.5*4)**	
	(a)What are the various ways of classifying the continuous time signal ?		
	(b) Evaluate the following integrals i) $\int_{-\infty}^{\infty} \delta(t+3)e^{-t}dt$ ii). $\int_{-\infty}^{\infty} \delta(t)cost + \delta(t-1)sintdt$		
	(c). What do you mean by Fourier transform? What are Dirichlet Conditions ?		
	(d). Determine the FT of the following signals i) $x(t) = e^{-3\|t\|}u(t)$ ii) $x(t)sin\Omega_o t$		
Q 2	**(Attempt any Two Parts)**	**(5,5)**	
	(a) Determine the range of values of 'a' and 'b' for the stability of LTI system with impulse response is given by h(t)= $e^{at}u(t) + e^{-bt}u(t)$		
	(b). Perform the linear convolution of the following signal x(n)={ 1,2 ,3,1} and h(n)={1,1,1} using graphical method		
	(C) Verify whether the following signals are periodic or non periodic signals i) $x(t) = 3cos\sqrt{2}t + 7cos5\pi t$ ii) $x(t) = 3cos(5t + \frac{\pi}{6})$		
Q 3	**(Attempt any Two Parts)**	**(5,5)**	
	(a). Determine the convolution of $x(t)=e^{-2t}u(t)$ and $h(t)= e^{-6t}u(t)$ using Fourier transform		
	(b). State and prove any two properties of Fourier transform.		
	(C). The impulse response of an LTI system is $h(t)=2e^{-3t}u(t)$.Find the response of the system for the input $x(t)=2e^{-5t}u(t)$ using Fourier transform.		